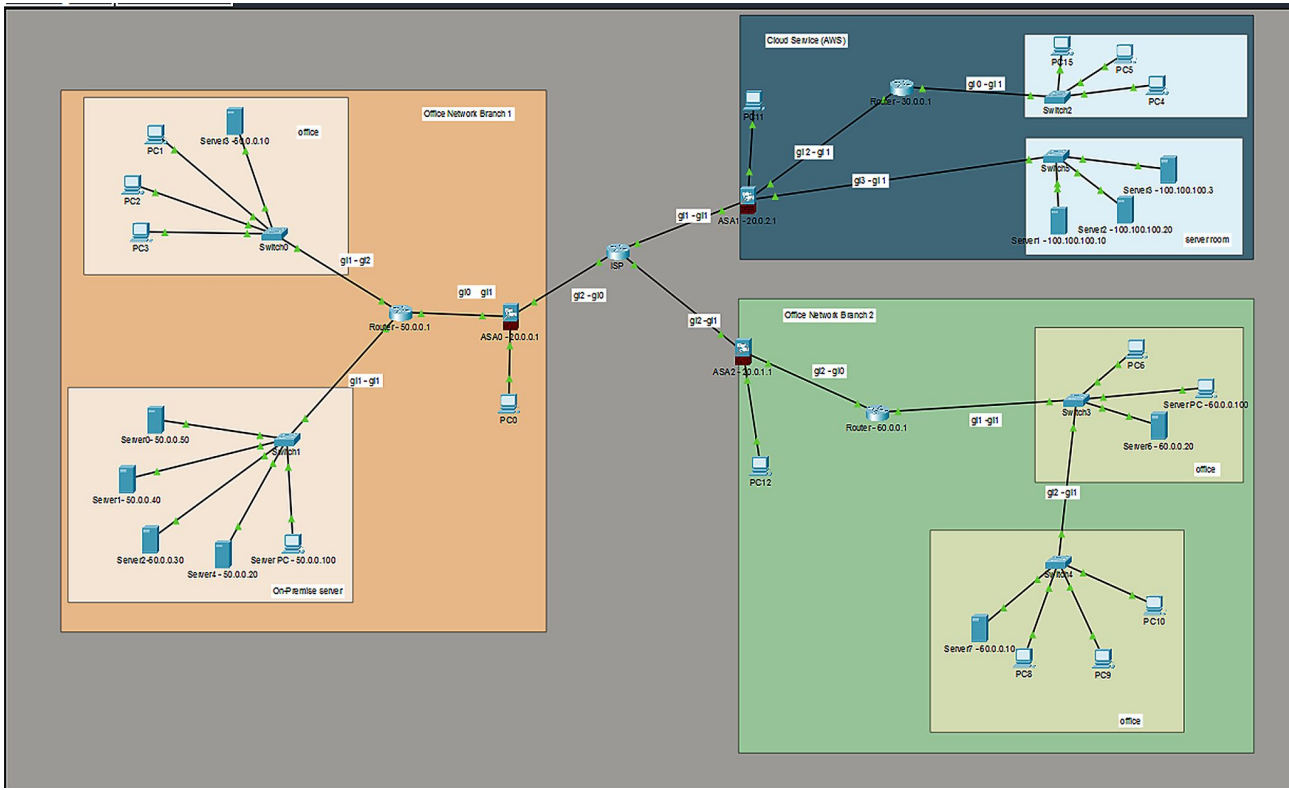
The Topology

Branch 1 and Branch 2 each sit behind their own ASA.

The cloud servers sit in a DMZ behind a third one.

A single ISP router joins all three.



The rules

- Both branches can reach the cloud servers
- Each branch can reach the other branch’s published servers
- The Server PC in each branch gets a VPN to the other branch
- Normal PCs cannot reach the other branch’s PCs
- In the cloud zone, clients can use the servers, but the servers cannot start connections to the clients

Addressing

Branch 1 LANs: 192.168.50.0/24 and 192.168.100.0/24

Branch 2 VLANs: 200.200.100.0/24 (VLAN 10) and 200.200.200.0/24 (VLAN 20)

Router to ASA links: 50.0.0.0/24 and 60.0.0.0/24

ASA to ISP: three /30 links (20.0.0.0, 20.0.1.0, 20.0.2.0)

Cloud: 30.0.0.0/29 for the client side, 100.100.100.0/24 for the DMZ servers

**These are lab addresses, not RFC 1918.

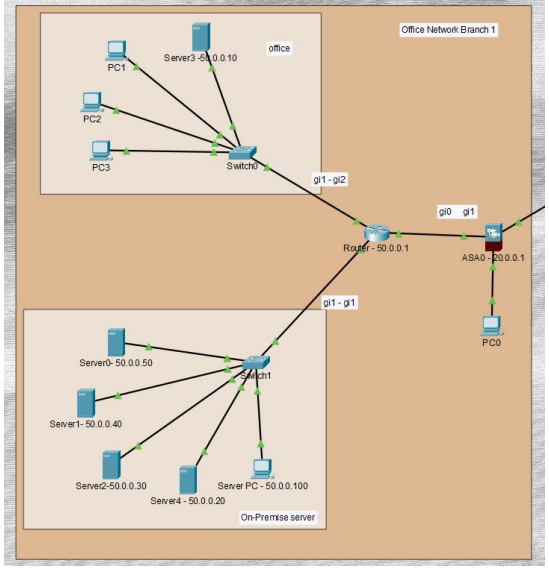
Branch routers

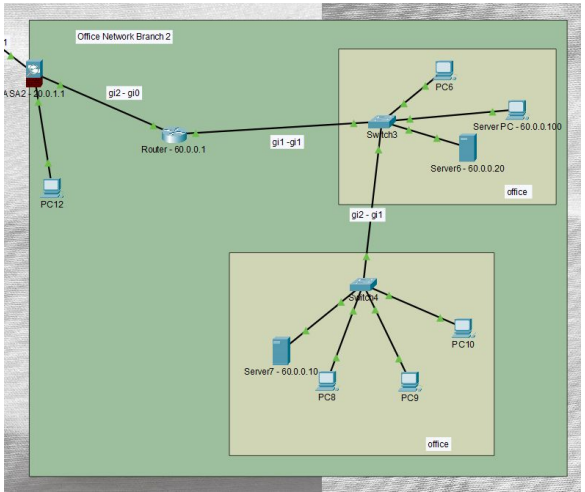
DHCP on every LAN, one pool each

Branch 2 uses router-on-a-stick for VLAN 10 and VLAN 20

PAT out of the router, plus static NAT for each server

Branch 1 has 6 static mappings, Branch 2 has 3





The ASAs

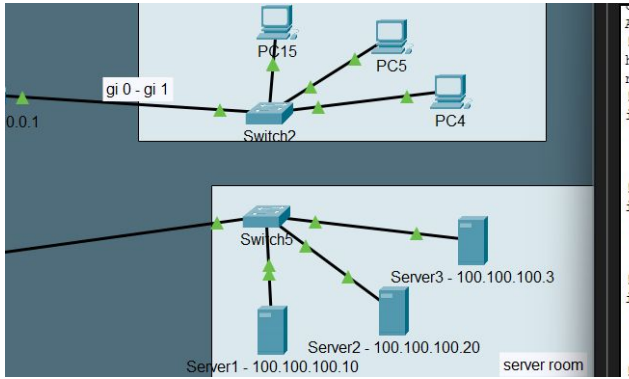
PAT for the whole inside network

Identity NAT for each published server, so it keeps its address

Branch ASAs open only FTP, HTTPS and RDP to the right servers

DMZ

- “Three zones on the cloud ASA: OUTSIDE (level 0), DMZ (level 50), INSIDE (level 100)”
- “Clients can reach the servers. The servers cannot start connections back to the clients”
- “Branches can reach only the DMZ servers, nothing else in the cloud zone”




```
ASA Version 9.6(1)
!
hostname ciscoasa
names
!
interface GigabitEthernet1/1
 nameif OUTSIDE
 security-level 0
 ip address 20.0.2.1 255.255.255.252
!
interface GigabitEthernet1/2
 nameif INSIDE
 security-level 100
 ip address 30.0.0.2 255.255.255.248
!
interface GigabitEthernet1/3
 nameif DMZ
 security-level 50
 ip address 100.100.100.1 255.255.255.0
!
```

- A DMZ on the cloud ASA, so the servers are kept apart from the

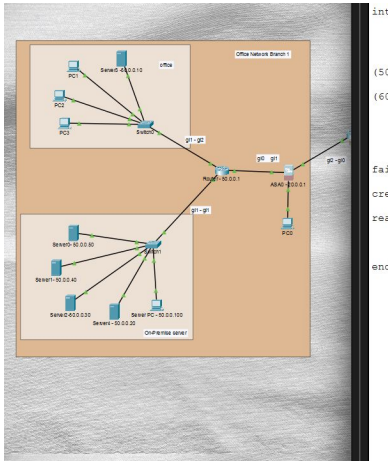
IPsec VPN

Tunnel between ASA0 and ASA2

IKEv1 with AES, a pre-shared key and DH group 5

ESP with AES-256 and SHA

A crypto ACL that matches only the traffic meant for the tunnel




```
interface: outside
    Crypto map tag: CMAP, seq num: 10, local addr 20.0.0.1

      permit ip object LOCALLAN object REMOTELAN
      local  ident (addr/mask/prot/port):
(50.0.0.0/255.255.255.0/0/0)
      remote  ident (addr/mask/prot/port):
(60.0.0.0/255.255.255.0/0/0)
      current_peer 20.0.1.1
      #pkts encaps: 5295, #pkts encrypt: 5295, #pkts digest: 0
      #pkts decaps: 9194, #pkts decrypt: 9194, #pkts verify: 0
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts comp failed: 0, #pkts decomp
failed: 0
      #pre-frag successes: 0, #pre-frag failures: 0, #fragments
created: 0
      #PMTUs sent: 0, #PMTUs rcvd: 0, #decapsulated frgs needing
reassembly: 0
      #send errors 1, #recv errors 0

      local crypto endpt.: 20.0.0.1/0, remote crypto
endpt.:20.0.1.1/0
      path mtu 1500, ip mtu, ipsec overhead 78,  media mtu 1500
      current outbound spi: 0x7B321C38(2066881592)
      current inbound spi: 0x7EA56673(2066881592)

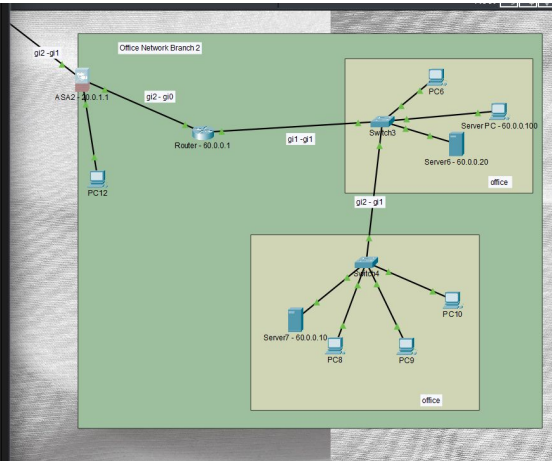
    inbound esp sas:
     spi: 0x7EA56673(2124768883)
        transform: esp-aes 256 esp-sha-hmac no compression
        in use settings ={L2L, Tunnel, }
        slot: 0, conn id: 2002, crypto map: CMAP
        sa timing: remaining key lifetime (k/sec): (4525504/83)
        IV size: 16 bytes
        replay detection support: N
        Anti replay bitmap:
         0x00000000 0x0000001F
    outbound esp sas:
```

```
interface: OUTSIDE
    Crypto map tag: CMAP, seq num: 10, local addr 20.0.1.1

      permit ip object LOCAL_LAN object REMOTE_LAN
      local  ident (addr/mask/prot/port):
(60.0.0.0/255.255.255.0/0/0)
      remote  ident (addr/mask/prot/port):
(50.0.0.0/255.255.255.0/0/0)
      current_peer 20.0.0.1
      #pkts encaps: 29, #pkts encrypt: 29, #pkts digest: 0
      #pkts decaps: 21, #pkts decrypt: 21, #pkts verify: 0
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts comp failed: 0, #pkts decomp
failed: 0
      #pre-frag successes: 0, #pre-frag failures: 0, #fragments
created: 0
      #PMTUs sent: 0, #PMTUs rcvd: 0, #decapsulated frgs needing
reassembly: 0
      #send errors 1, #recv errors 0

      local crypto endpt.: 20.0.1.1/0, remote crypto
endpt.:20.0.0.1/0
      path mtu 1500, ip mtu, ipsec overhead 78,  media mtu 1500
      current outbound spi: 0x0531BBF6(87145462)
      current inbound spi: 0xF96FC702(87145462)

    inbound esp sas:
     spi: 0xF96FC702(4184852226)
        transform: esp-aes 256 esp-sha-hmac no compression
        in use settings ={L2L, Tunnel, }
        slot: 0, conn id: 2002, crypto map: CMAP
        sa timing: remaining key lifetime (k/sec): (4525504/3451)
        IV size: 16 bytes
        replay detection support: N
        Anti replay bitmap:
         0x00000000 0x0000001F
    outbound esp sas:
```



ISP router

Only three static routes:

50.0.0.0/24 via 20.0.0.1

60.0.0.0/24 via 20.0.1.1

100.100.100.0/24 via 20.0.2.1

Because of the double NAT, the ISP never sees a 192.168.x.x or 200.200.x.x address.

What broke

1. VPN traffic reached the ASA and got dropped. Packet Tracer wanted it permitted on the outside ACL.
2. The first ping always timed out. It was ARP, so I pinged the peer once before testing to cache ARP.
3. Object network group is not support on Packer Tracer ,so created objects for each identity NAT.

The files

GitHub: https://github.com/MoorthyRJ/Enterprise_pkt.git

Drive (.pkt file): https://drive.google.com/drive/folders/1dbFc0TPYua6pXBIzlctz5V4wbXUJLpQa?usp=drive_link

Open the .pkt in Packet Tracer 9.0.0 or newer.